

ECON-1020H Introductory Macroeconomics

Chapter 5 Measuring National Income

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2025-09-15

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Fall 2025

National accounts

Gross Domestic Product

Calculating GDP

Real versus Nominal GDP

GDP and Economic Well-Being

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- This week: Textbook Chapter 5
- Next week: Textbook Chapter 6

Introduction

- Economic data at the economy level, i.e., macroeconomic indicators, are needed
 - to track the condition of the economy, and
 - to study the macroeconomy
- Macroeconomic indicators are not readily available
- Economists must create those indicators. This course focuses on
 - Aggregate income and output
 - Inflation
 - Unemployment
- This chapter studies aggregate income

National accounts

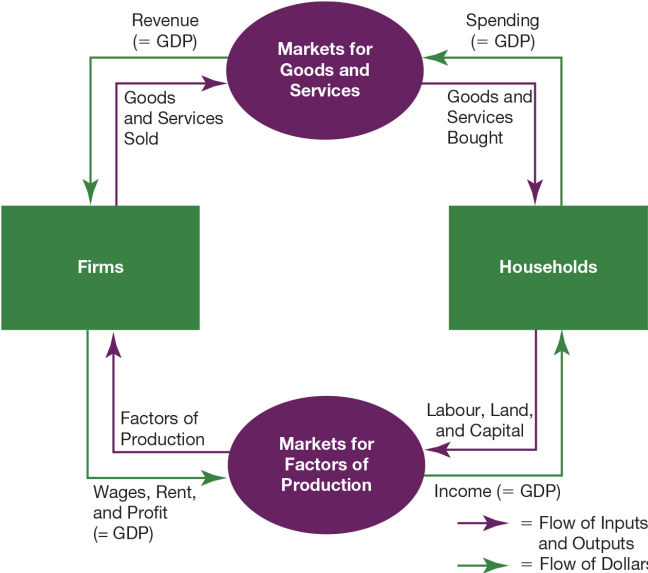
National accounts

National accounts are tables that record income and spending at the national level, at provincial level, and by sector

Every nation has agencies building and updating national accounts:

- In Canada, **Statistics Canada**
 - **National Gross Domestic Product (GDP) by Income and by Expenditure Accounts**
- In the U.S., **Bureau of Labor Statistics** and **the Census Bureau**

Circular-flow diagram



Circular-flow diagram is useful in counting expenditures and incomes

For an economy as a whole, income must equal expenditure

- A person's spending is another person's income
- Thus, at the aggregate level, income is expenditure
- It is two ways of calculating the same aggregate income

The measure of aggregate income is Gross Domestic Product (GDP)

Gross Domestic Product

Defining GDP

Gross domestic product (GDP) is one key variable measuring the macroeconomy

What is GDP?

Gross domestic product is the total market value of all final goods and services produced within a country in a given period of time

The definition above shows what are included in GDP and what are not

What are measured in GDP, and what are not

... total market value ...

- GDP adds different kinds of products into a single measure of the value of economic activity
- GDP measures the value of goods and services for which there are markets (hence prices)
- Goods and services that have no market prices (hence not in GDP):
 - Home production, e.g., child care at home, cooking at home
 - "Free" stuff, e.g., YouTube, Facebook, Gmail, ...
 - Illegal activities, e.g., illicit drugs
 - Environment quality
- Housing services provided by owner-occupied dwelling is in GDP
 - Imputed rent

... final goods and services ...

GDP includes the value of final goods, because the value of intermediate goods is already included in the prices of final goods

E.g., furniture is a final good, lumber used in making furniture is an intermediate good. Furniture price includes the cost of lumber. Thus furniture is measured as part of GDP, but lumber (used for making furniture) is not

But, a firm's inventory of good for use or sale at a later date is included as part of GDP

Which goods and services are final?

- Consumption spending by households
- Investment spending by households and businesses
- Government purchases
- Exports (goods bought by foreigners)

Which goods and services are intermediate? Those "consumed" in production

- Raw materials, services used in production, and energy used in production

Examples of final and intermediate goods:

- Traveling services used for business is intermediate, traveling services purchased for your vacation is final
- Electricity used in production is intermediate, electricity for your EV is final
- My expenses of commuting to work are final, but is intermediate if it is reimbursed by my employer
- New machines used in production is final, not intermediate

“... produced within a country in a given period of time...”

GDP measures the value of production within the geographic confines of a country

- Table lamp produced by your factory in the U.S. is in the U.S. GDP, not in the Canadian GDP
- The k-pop album imported from South Korea is not in Canadian GDP

GDP measures values of goods and services that are currently produced

- The value of a used car is not in GDP, it was in GDP when first produced

GDP measures production in a year, a quarter, or a month

- Quarterly GDP growth **at annual rate** is quarterly growth multiplied by 4

GDP measures two things at once

In circular-flow diagram, **total expenditure equals total factor income**

- Factors of production includes labor, capital, and entrepreneurship

GDP measures the **total factor income** in an economy

Total factor income (also called aggregate income) is the sum of incomes of factors of production, including

- labor income,
- capital income,
- profit income, and
- net taxes paid by producers

GDP also measures the **total expenditure** on final goods and services

- **Aggregate expenditure** consists of
 - consumption expenditure C
 - investment spending I
 - government expenditure G
 - Net exports (NX): exports X subtracting imports M

The income-expenditure identity

Total income equals total expenditure

Let Y be GDP as aggregate income

- $Y = \text{Labor income} + \text{Capital income} + \text{Profits}$

Income-expenditure Identity

$$Y = C + I + G + X - M.$$

- Denote $(X-M)$ as NX , net exports

Is saving in GDP?

In later chapters, we will discuss saving

For now, saving is part of income, hence in GDP

Let S be saving, then $S = Y - C - G$ for the economy

In open economy, $S = (C + I + G + X - M) - C - G = I + X - M$

In a closed economy, $S = I$

Calculating GDP

Calculating GDP

According to the definition, three methods of measuring GDP

- GDP can be measured as aggregate income
- GDP can be measured as aggregate expenditure
- GDP can be measured as total value added (i.e., revenue minus costs of intermediate goods)

Three measures lead to the same dollar value of GDP

We focus on the expenditure approach and the income approach

GDP Components (by the expenditure approach)

Consumption C is spending by households on goods and services, excluding purchases of new housing

Examples:

- Foods, utility bills, haircuts, etc
- Imputed rents from services provided by a owner-occupied home
- Spending on education
- Spending on imported consumption goods, e.g., imported fruits and clothes

Consumption spending on both domestic and imported goods and services

Investment I is the purchase of goods used for production in the future

It includes the purchases of:

- business capital, i.e., equipment, structures, land, and intellectual property products (e.g. R&D, software)
 - including machines and equipment imported from abroad
- Residential capital, i.e., landlords' apartment building and households' residence
- Inventories, i.e., unused materials, unsold products.

Government purchases G measures spending on goods and services by local, provincial, and federal governments. It includes

- salaries of government employees,
- expenditures on public works, and
- investment in capital

Calling investment in capital a government purchase is a simplification

Transfer payments, e.g., social security benefit, unemployment insurance benefit, are NOT part of G ,

- because these payments are not made in exchange for a currently produced good or service

Transfer payments are part of household income, but they do not reflect production

Net Exports NX are exports minus imports

Examples of exports and imports:

- (Ask class)

Why do we subtract imports from GDP?

- Imported goods are used for consumption and investment
- But they are not produced in the importing country

Why should we subtract imported goods used as intermediate inputs?

GDP measures market values of final goods and services

The value of final goods and services has two components:

- Value of intermediate goods (e.g., raw materials)
- Value added to intermediate goods

By counting only the value of final goods, GDP already contains the value of (expenditure on) intermediate goods

But, imported intermediate goods are not produced within the country, thus their values should be subtracted from GDP

(show an example)

An example

A hypothetical economy with two producers, *A* and *B*, and many consumers

Firm *B* produces rubber, steel, as well as furniture

In addition to labor and capital,

- producing rubber requires natural rubber, an abundant endowment
- producing steel needs iron ores, abundant in the country
- producing furniture requires imported woods

Firm *A* produces cars, both are made of rubber and steel purchased from firm *B*

Revenues and expenses at two firms in 2022, in millions dollars

Firm A			Firm B		
Revenue from domestic sales		1000	Revenue from domestic sales		570
Revenue from exports		120	Revenue from exports		0
Expenses, total		1100	Expenses, total		560
Expenses, details			Expenses, details		
Wages	330		Wages	264	
Capital rental	220		Capital rental	216	
Rubber and steel	550		Wood imports	80	
Profit		20	Profit		10

By the expenditure approach, GDP in the example economy is 1,060 million dollars

- Consumption spending is 1,020 million dollars ($=1000+20$)
- Investment spending is zero
- Exports are 120 million dollars
- Imports are 80 million dollars

Ever wonder why "gross" domestic product?

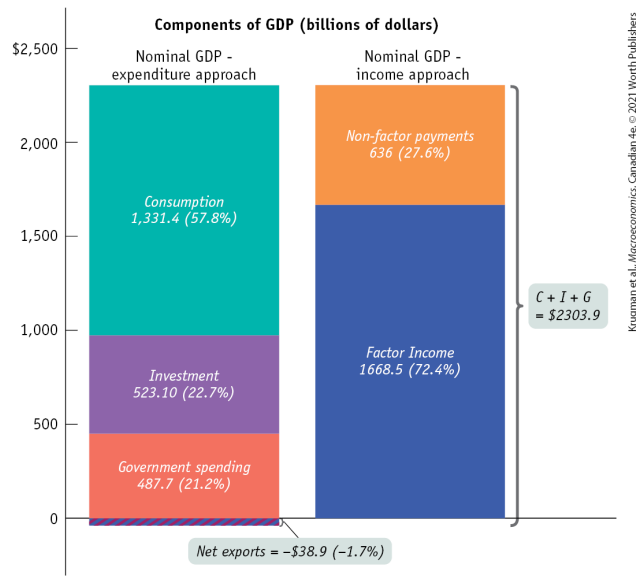
The word "gross" in GDP is because **depreciated capital** is not subtracted from GDP

- Depreciation refers to wears and tears of machines in production
- Recall, things that are consumed in production are intermediate goods
- Wears and tears are the part of capital that is consumed in production

So, we should subtract the value of capital lost in production

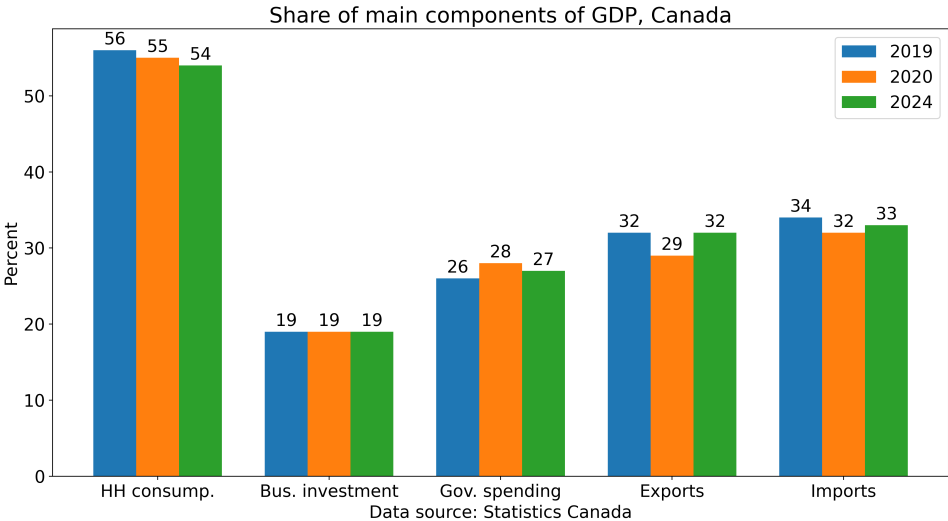
But, in practice, we don't, therefore the name "gross" domestic product

Aggregate income and aggregate expenditure in Canada (2019)



Krugman et al., Macroeconomics, Canadian 4e, © 2021 Worth Publishers

Components of gross domestic product in Canada



Real versus Nominal GDP

Nominal GDP versus Real GDP

Nominal GDP is the value of final goods and services produced during a given year valued at prices prevailed in the same year

Real GDP is the value of final goods and services produced in a given year when valued at prices prevailed in a base year

- Thus, real GDP must be said together with the chosen base year

By valuing current production using past prices, real GDP shows how the economy's production of goods and services changes over time

Why real GDP?

- Real GDP measures the quantity of aggregate output
- It is the quantity not current dollar value that matters for well-being
- If current dollar value increases because only prices rise, that increase does not change the well-being (welfare)

A simple example of calculating real GDP

Item		Quantity (millions)	Price (dollars)	Expenditure (millions of dollars)
(a) In 2007				
C	T-shirts	10	5	50
I	Computer chips	3	10	30
G	Security services	1	20	20
Y	Real GDP in 2007			100
(b) In 2016				
C	T-shirts	4	5	20
I	Computer chips	2	20	40
G	Security services	6	40	240
Y	Nominal GDP in 2016			300
(c) Quantities of 2016 valued at prices of 2007				
C	T-shirts	4	5	20
I	Computer chips	2	10	20
G	Security services	6	20	120
Y	Real GDP in 2016			160

Example from textbook

Prices and Quantities				
Year	Price of Hot Dog	Quantity of Hot Dogs	Price of Hamburger	Quantity of Hamburgers
2020	\$1	100	\$2	50
2021	2	150	3	100
2022	3	200	4	150
Year	Calculating Nominal GDP			
2020	(\$1 per hot dog $\times$ 100 hot dogs) + (\$2 per hamburger $\times$ 50 hamburgers) = \$200			
2021	(\$2 per hot dog $\times$ 150 hot dogs) + (\$3 per hamburger $\times$ 100 hamburgers) = \$600			
2022	(\$3 per hot dog $\times$ 200 hot dogs) + (\$4 per hamburger $\times$ 150 hamburgers) = \$1200			
Year	Calculating Real GDP (base year 2020)			
2020	(\$1 per hot dog $\times$ 100 hot dogs) + (\$2 per hamburger $\times$ 50 hamburgers) = \$200			
2021	(\$1 per hot dog $\times$ 150 hot dogs) + (\$2 per hamburger $\times$ 100 hamburgers) = \$350			
2022	(\$1 per hot dog $\times$ 200 hot dogs) + (\$2 per hamburger $\times$ 150 hamburgers) = \$500			
Year	Calculating the GDP Deflator			
2020	$(\$200/\$200) \times 100 = 100$			
2021	$(\$600/\$350) \times 100 = 171$			
2022	$(\$1200/\$500) \times 100 = 240$			

GDP deflator is a measure of the price level

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \cdot 100$$

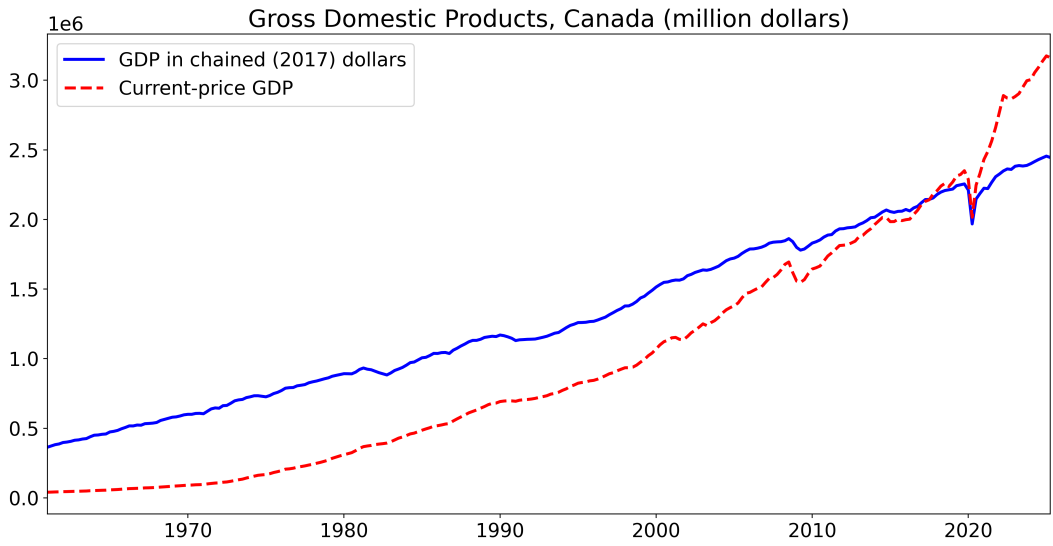
$$\text{In the example above, GDP deflator in 2022} = \frac{1200}{500} \times 100 = 240$$

$$\text{To better understand it, GDP deflator in 2022} = \frac{3 \cdot 200 + 4 \cdot 150}{1 \cdot 200 + 2 \cdot 150} \cdot 100$$

The inflation rate of GDP deflator is the percentage change in GDP deflator

$$\text{E.g., Inflation rate in 2022} = \frac{240 - 171}{171} \cdot 100\% = 40.35\%$$

Gross domestic products in Canada



Data source: Statistics Canada

GDP and Economic Well-Being

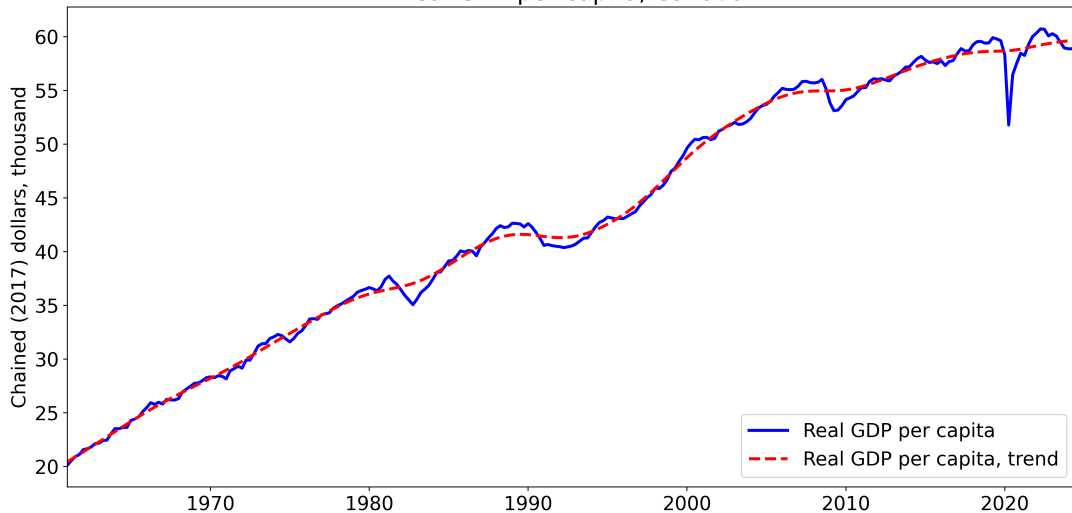
GDP as a measure of well-being

Real GDP per capita is real GDP divided by the population, it is a measure of the standard of living

Real GDP per person in Canada, since 1971

- 1971: 28,869 dollars in 2017 prices
- 2019: 59,681 dollars in 2017 prices
- An increase of 107 percent from 1971 to 2019
- Fast growth in 1990s
- Slower growth since 2000s, and since 2009
- 2023: 58,945 dollars in 2017 prices

Real GDP per capita, Canada



Data source: Statistics Canada

Much is left out of GDP

Recall, GDP excludes the value of activities taking place outside markets

More things that are not measured in GDP:

- Value of leisure
- Investment in human capital
- GDP says little about the distribution of income

Thus, GDP per capita is an imperfect measure of well-being

Appendix: Alice and Paul's contribution to GDP

In each event, is it in GDP?

- In 2019, Paul rented basement in Alice's house (rent). Alice hired Paul to maintain backyard (service). Paul went to Gym regularly (service)
- In 2020, Paul and Alice were married, they live in the same house (imputed rent). Paul still maintained backyard (home production). Alice, a data scientist, prepared family dinner (home production). She received CERB (government transfer)

- In 2021, Alice gave birth to a baby girl. Paul now worked out at home as gym was closed (home production?)
- In 2022, Alice is back to work (labor income). Paul is now a trainer and a Youtuber, sharing his work-out routines (self employed? Gig economy?). Paul also takes care of their baby girl (home production) when their baby-sitter cannot work. (Back to [main text](#))